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1604 boxes, dem, nodes = data["boxes"], data["dem"], data["nodes"]
1605 depot = nodes["001"]
1606 units_by_type: dict[str, list[str]] = {
1607     type_id: [u["unit_id"] for u in data["transport_units"] if u["type_id"] == type_id]
1608     for type_id in data["transport_types"]
1609 }
1610 battery_ids = {
1611     type_id: [f"{type_id}-BAT-{i:02d}" for i in range(1, info["count"] + 1)]
1612     for type_id, info in data["battery_inventory"].items()
1613 }
1614 unit_ready = {u["unit_id"]: 0.0 for u in data["transport_units"]}
1615 battery_ready = {battery: 0.0 for values in battery_ids.values() for battery in values}
1616 scheduled: list[dict[str, Any]] = []
1617
1618 def options(spec: dict[str, Any], forced_unit: str | None = None) -> list[tuple[Any, ...]]:
1619     candidates = []
1620     for type_id, g in data["transport_types"].items():
1621         units = [forced_unit] if forced_unit in units_by_type[type_id] else units_by_type[type_id]
1622         if forced_unit is not None and forced_unit not in units_by_type[type_id]:
1623             continue
1624         for unit_id in units:
1625             for battery_id in battery_ids[type_id]:
1626                 start = max(unit_ready[unit_id], battery_ready[battery_id])
1627                 timeline = route_timeline(dem, depot, nodes, g, boxes, spec["stops"], start)
1628                 if timeline is None:
1629                     continue
1630                 hard_late = sum(not x["hard_on_time"] for x in timeline["completions"])
1631                 hard_max = max(
1632                     (max(0.0, x["completion_s"] - x["hard_deadline_s"])
1633                      for x in timeline["completions"] if x["hard_deadline_s"] is not None),
1634                     default=0.0,
1635                 )
1636                 wtd = sum(
1637                     x["priority"] * x["tardiness_s"] / x["expected_s"]
1638                     for x in timeline["completions"] if x["hard_deadline_s"] is not None
1639                 )
1640                 score = (hard_late, hard_max, wtd, max(x["completion_s"] for x in timeline["completions"]),
1641                         timeline["energy_kwh"], timeline["return_s"], type_id, unit_id, battery_id)
1642                 candidates.append((score, type_id, unit_id, battery_id, timeline))
1643     return candidates
1644
1645 urgent = [x for x in specs if _trip_spec_key(x, boxes)[0] == 3600.0]
1646 remaining = [x for x in specs if x not in urgent]
1647 all_units = [u["unit_id"] for u in data["transport_units"]]
1648 if urgent and all(x.get("forced_unit") for x in urgent):
1649     initial_choices = [
1650         min(options(spec, forced_unit=spec["forced_unit"]), key=lambda x: x[0])
1651         for spec in urgent
1652     ]
1653 elif len(urgent) == len(all_units):
1654     pair_options: dict[tuple[int, str], tuple[Any, ...]] = {}
1655     for job_index, spec in enumerate(urgent):
1656         for unit_id in all_units:
1657             cands = options(spec, forced_unit=unit_id)
1658             if cands:
1659                 pair_options[(job_index, unit_id)] = min(cands, key=lambda x: x[0])
1660     best_assignment = None
1661     for perm in itertools.permutations(all_units):
1662         chosen = [pair_options.get((i, perm[i])) for i in range(len(urgent))]
1663         if any(x is None for x in chosen):
1664             continue
1665         assert all(x is not None for x in chosen)
1666         score = (
1667             max(x[0][1] for x in chosen),
1668             sum(x[0][0] for x in chosen),
1669             max(x[4]["completions"][-1]["completion_s"] for x in chosen),
1670             sum(x[4]["energy_kwh"] for x in chosen),
1671             perm,
1672         )
1673         if best_assignment is None or score < best_assignment[0]:
1674             best_assignment = (score, chosen)
1675     if best_assignment is None:
1676         raise RuntimeError("No feasible assignment for eight 3600-second hard trips")
1677     initial_choices = best_assignment[1]
1678 else:
1679     initial_choices = [min(options(spec), key=lambda x: x[0]) for spec in urgent]
1680

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